7 **(a)** Simple harmonic motion is an oscillatory motion whereby the acceleration of a body is directly proportional to its displacement from the equilibrium position, and the direction of the acceleration always acts towards the equilibrium position.

(b) (i) $f = \dfrac{1}{T}$

$f = \dfrac{1}{0.6} \approx 1.7\ \text{Hz}$

(ii) For simple harmonic motion, the velocity of the plate is $v = \pm\omega\sqrt{x_o^{\ 2} - x^2}$ where x_o is the amplitude and x is the displacement.

At $x = 0$, the plate experiences maximum velocity and has maximum kinetic energy. Hence, we have

$$E_{K_{\max}} = \frac{1}{2} m v_{\max}^{\ 2}$$
$$E_{K_{\max}} = \frac{1}{2} m\omega^2 \left(x_o^{\ 2} - 0\right)$$
$$E_{K_{\max}} = \frac{1}{2} m\omega^2 a^2$$

At this position, the total energy $E_T = E_{K_{\max}} = \dfrac{1}{2} m\omega^2 a^2$ since potential energy is zero.

Hence, we have $E_T = \dfrac{1}{2} m\left(2\pi f\right)^2 a^2 = 2\pi^2 m f^2 a^2$

Note: Candidates should explain that the total energy of the oscillation is equal to the maximum kinetic energy.

(iii) $E_T = 2\pi^2 m f^2 a^2$

$E_T = 2\pi^2 \times 320 \times 10^{-3} \times 1.67^2 \times \left(10 \times 10^{-3}\right)^2$

$E_T \approx 1.8 \times 10^{-3}\ \text{J}$

(c) (i) The damping of the plate is light since the amplitude decreases by a very small amount as time progresses, with the period remaining constant.

(ii) 1. Faraday's law of electromagnetic induction states that the induced e.m.f. in a circuit is directly proportional to the rate of change of magnetic flux linkage through the circuit.

2. When the current is switched on, there is a magnetic field created within the coil that cuts through the plate. As the plate oscillates, it cuts the magnetic field and experiences a change in magnetic flux. By Faraday's law, there is an induced e.m.f. in the plate. By Lenz's law, the induced e.m.f. will produce effects to oppose the change in flux. In this case, the plate set up induced currents, which causes itself to experience opposing magnetic force against its motion. Hence, the oscillations are damped.

(iii) After 3.6 s, the amplitude of oscillations is 5 mm.

Loss of energy

$$= 2\pi^2 \times 320 \times 10^{-3} \times 1.67^2 \times \left[\left(10 \times 10^{-3}\right)^2 - \left(5 \times 10^{-3}\right)^2 \right]$$

$$\approx 1.32 \times 10^{-3}\,\text{J}$$

(d) The introduction of the iron core increases the magnetic field strength within the coil since soft iron is a ferromagnetic material. The induced e.m.f. in the plate will increase since the plate now experiences a greater change in flux due to the increased in magnetic field that cuts through it. Hence, the induced current in the plate increases, which causes the opposing magnetic force to increase. The oscillations will be affected as there will be much more damping than before.

8 **(a)** Resistance *R* is a property of a sample (or electrical component) where it is defined